

Il Young Jeong

william04020@unist.ac.kr — github.com/Al3ph-Zer0 — al3ph-zer0.github.io

Education

Ulsan National Institute of Science and Technology (UNIST)

Ulsan, Korea

M.S. student, Mathematical Sciences

March 2026–Present

- Research focus: model theory and asymptotic behavior of classes of models.
- Advised by : Junguk Lee, Hyeungjoon Kim

Ulsan National Institute of Science and Technology (UNIST)

Ulsan, Korea

B.S., Mathematical Sciences

February 2026

- Relevant coursework: Algebra, Linear Algebra, Real Analysis, Probability, Differential Equations, Logic.
- Chair of Club Union Bylaws Amendment Committee (2024 -)
- Manager of 'NeulPum' Election Camp for the Club Association (2025)
- Director of Academic Affairs of Mathematical Sciences (2025)
- Independent Delegate of Student Committee of College of Natural Sciences (2025)
- Vice of Student Committee of College of Natural Sciences (2024)
- Chair of Student Committee of College of Natural Sciences Bylaws Amendment Committee (2024)
- President of Club Union Academic Division (2024)
- Secretary of EOE, Math & Physics Club (2026)
- President of EOE, Math & Physics Club (2024)
- Chair of Math Division of EOE, Math & Physics Club (2023)
- President of ULECA, Chemistry Club (Feb - Aug 2024)
- Vice of ULECA, Chemistry Club (Sep 2024 - Jan 2025)

Research Interests

- Model theory.
- Zero-one laws and zero-one properties for classes of finite structures and models.
- Asymptotic behavior of classes of models, especially from a logical and model-theoretic viewpoint.
- Interactions between model theory, finite model theory, and algebraic examples.

Projects and Notes

Asymptotic behavior of classes of models

In progress

- Studying zero-one phenomena and asymptotic behavior for classes of models.
- Preparing notes and beamer presentations related to model-theoretic asymptotics.

Math study archive

al3ph-zer0.github.io

- Maintains a personal archive for CV materials and mathematical study notes.
- Planned sections include model theory, algebra, analysis, and reading notes.

Presentations and Academic Activities

- **“Fraïssé Classes and the First-Order Limit Property”**, invited student talk, The Fifth Korea Logic Day 2026, KAIST, Daejeon, January 14, 2026.
- **Model Theory Workshop**, lecturer and organizer, UNIST, 2026.
- **Talk on asymptotic behavior of classes of models and zero-one phenomena**, Korean Mathematical Society 80th Anniversary Meeting, 2026.

Skills

Mathematics	Model theory, logic, finite structures, algebra, analysis.
Programming	Python, LaTeX, Lean.
Languages	Korean; English
Writing	To be added (soon maybe).

Selected Coursework

- Algebra I